On Integrated Information

2026-03-23

The Sovereignty Papers

Essay No. 4: On Integrated Information

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> "Information is a difference that makes a difference."
>
> — Gregory Bateson, *Steps to an Ecology of Mind* (1972)

I. Two Committees

Imagine two governance systems.

Both are deciding water policy for the same community.

In the first, a committee of five members votes on water policy. Each member has read the relevant reports. Each has attended public hearings. Each has reviewed the environmental data, consulted with affected communities, and studied the engineering constraints. When they deliberate, each member's understanding is informed by every other dimension of the problem — the environmental data changes how they interpret the community feedback, the engineering constraints shape how they evaluate the environmental risks, the community input modifies their assessment of which engineering tradeoffs are acceptable. The five members are not merely aggregating separate inputs. They are *integrating* them — each dimension of their understanding shapes every other dimension.

In the second, a committee of five members votes on the same water policy. Each has read different reports. None has consulted with the others. They arrive at the vote with five independent assessments, cast five independent ballots, and the majority wins. The outcome may be identical to the first committee's. But the process is fundamentally different: the five assessments were never integrated. No member's understanding was shaped by any other member's knowledge. The system aggregated inputs. It did not integrate them.

The first committee is, in the operational sense we defined in Essay No. 3, *conscious* of the water policy domain. The second is not — even though it may contain the same information in aggregate. The difference is integration: whether the parts of the system causally influence each other in a way that produces a unified understanding, or whether they merely coexist.

This distinction — between genuine integration and mere aggregation — is the question on which the entire project of consciousness-coupled governance stands or falls. If we cannot measure it, then the thesis of Essay No. 3 is an aspiration, not an architecture. The history of governance is a history of measurement failures: the 1602 architecture measures capital and calls it voice; democracy measures citizenship and calls it consent; meritocracy measures credentials and calls it competence. Each captures something real but excludes something essential — the integration of a participant's understanding across all relevant dimensions.

The distinction between integration and aggregation is the core insight of Integrated Information Theory, developed by the neuroscientist Giulio Tononi beginning in 2004.[1]

Tononi's insight was that consciousness is not about the amount of information a system contains but about how much information is generated by the system *as a whole* beyond what is generated by its parts independently. A brain is conscious not because it contains billions of neurons but because those neurons are connected in ways that make the whole brain generate more information than any partition of it into independent subsets. A hard drive may contain more raw information than a brain, but its bits do not causally influence each other — each bit can be read independently of every other bit. The hard drive has information. The brain has *integrated* information.

Tononi called his measure Phi (Φ), from the Greek letter conventionally used for the golden ratio — a measure of how much a system is "more than the sum of its parts."[2]

III. Why This Matters for Governance

The reader who has followed the argument from Essay No. 1 will immediately recognize the governance parallel.

The VOC's Heeren XVII were an *aggregation* system, not an *integration* system. Each director brought information from his chamber — Amsterdam, Middelburg, Delft, Rotterdam, Hoorn, Enkhuizen. But the architecture did not require that each director's understanding be shaped by every other director's knowledge, let alone by the experience of affected populations. The directors voted. The majority won. The system aggregated preferences. It did not integrate understanding.

This is precisely the architecture of token-weighted governance in DAOs. Each token holder

[1] Giulio Tononi, "An Information Integration Theory of Consciousness," *BMC Neuroscience* 5, no. 42 (2004). The theory has been substantially developed since its initial formulation; see Tononi, "Integrated Information Theory of Consciousness: An Updated Account," *Archives Italiennes de Biologie* 150 (2012), and Tononi et al., "Integrated Information Theory: From Consciousness to Its Physical Substrate," *Nature Reviews Neuroscience* 17, no. 7 (2016).

[2] Tononi's choice of the letter Phi (Φ) is deliberate but his own — it does not derive from the golden ratio. The symbol was chosen to represent the amount of integrated information generated by a system above and beyond the information generated by its parts. In IIT 3.0 and later formulations, Phi is defined as the minimum information distance between a system's cause-effect structure and its minimum information partition.

brings their own assessment — informed by their financial position, their market analysis, their personal interests. They vote. The tokens are tallied. The system aggregates capital-weighted preferences. Nothing in the architecture requires, incentivizes, or even permits the kind of cross-domain integration that characterizes a genuinely conscious decision.

And this is why DAO governance produces the outcomes documented in Essay No. 2. MakerDAO's Black Thursday response was an aggregation of large holders' preferences, not an integration of large holders' financial interests with small vault holders' losses with the protocol's long-term stability. Compound's four-day timelock was an aggregation process — vote, wait, execute — disconnected from the consequence timeline. Beanstalk was a pure aggregation exploit: the attacker acquired enough units of the aggregation metric (tokens) to control the output, without any integration with the system's purpose, community, or consequences.

Phi measures exactly the property these systems lack. A governance system with high Phi is one in which the participants' understanding is genuinely integrated — where each participant's assessment is causally shaped by every relevant dimension of the problem, and where the collective decision reflects more understanding than any subset of participants could produce independently. A governance system with low Phi is one in which participants vote independently based on separate, unintegrated assessments — regardless of how many participants there are or how much information they individually possess.

The Heeren XVII had low Phi. They had information. They had power. They did not have integration. The Banda massacre was a low-Phi governance decision: the military assessment, the commercial assessment, and the human assessment were never integrated into a unified understanding. Each was processed in isolation, and the commercial assessment won because it was the only one connected to the reward function.

IV. The Honest Limitations

We have explained what Phi measures and why it matters for governance. Now we must be honest about what it cannot do.

Phi, as formally defined by Tononi, is computationally intractable for any system of significant size. Computing exact Phi requires evaluating every possible partition of a system into two or more subsets and measuring the information generated by each partition compared to the whole. For a system with n elements, the number of possible partitions grows super-exponentially. For a human brain with 86 billion neurons, exact Phi is not merely difficult to compute — it is physically impossible to compute in the lifetime of the universe.[3]

This is a real limitation, not a technicality. Any claim that a governance system "measures Phi" and uses it to assign governance power must reckon with the fact that what is being

[3] The computational intractability of exact Phi is well-established. See Tegmark, "Consciousness as a State of Matter," *Chaos, Solitons & Fractals* 76 (2015), for an analysis of the computational complexity. For a system of n binary elements, the number of possible partitions is given by the Bell number B(n), which grows faster than exponential. For n = 86 billion (approximate number of neurons in the human brain), exact Phi computation is not feasible by any known algorithm.

measured is not exact Phi but a proxy — an approximation that captures some aspects of integration while necessarily missing others.

We do not hide from this limitation. We face it directly, because facing it is the only way to build a system that can be trusted.

Symthaea uses proxy measures of integration, not exact Phi. The primary proxy is spectral connectivity — specifically, the Fiedler value of the system's causal graph, which measures how tightly connected the graph is.[4] A high Fiedler value indicates that the system cannot be easily partitioned into disconnected subsets — that information flows between all parts of the system. A low Fiedler value indicates that the system can be split into weakly connected components — that some parts of the system are isolated from others.

This is not Phi. It is a proxy for one of Phi's key properties — irreducible connectivity. It can be computed in polynomial time, which makes it practical for a real-time governance system. But it captures only one dimension of integration, and it can, in principle, be gamed by a system that creates artificial connections between its components without genuinely integrating the information they contain.

We address this risk in three ways.

First, the proxy is not used alone. Consciousness coupling, as defined in Essay No. 3, is a four-dimensional composite: presence, accountability, embeddedness, and engagement. Spectral connectivity contributes to the engagement dimension — it measures whether a participant's interactions with the governed domain form an integrated pattern or a disconnected set of activities. But it is combined with three other dimensions that are independently measured and independently verifiable. Gaming the spectral proxy would increase one dimension of the composite score but would not affect the other three.

Second, the credential expires. Every 24 hours, the consciousness credential must be recomputed from current data. A participant who artificially inflated their connectivity score yesterday must do so again today. Static gaming is ineffective because the measurement never settles into a permanent entitlement. This is the temporal coherence principle from Essay No. 1 applied to the measurement itself: just as governance decisions must be responsive to current consequences, governance credentials must be responsive to current engagement.[5]

Third, and most importantly: **the standard is not perfection. The standard is the 1602 architecture.**

[4] The Fiedler value — the second-smallest eigenvalue of the graph Laplacian — measures the algebraic connectivity of a graph. Named after Miroslav Fiedler, who introduced it in "Algebraic Connectivity of Graphs," *Czechoslovak Mathematical Journal* 23, no. 2 (1973). A graph with a high Fiedler value is tightly connected and difficult to partition; a graph with a low Fiedler value can be easily separated into weakly connected components. In the context of consciousness measurement, the causal interaction graph of a system's components is analyzed — a high Fiedler value indicates strong integration across all components.

[5] The 24-hour TTL on consciousness credentials was chosen as a balance between measurement freshness and practical overhead. In principle, the expiry period should match the consequence timescale of the governed domain — faster-moving domains (emergency response, financial protocols) might warrant shorter TTLs, while slower-moving domains (land use planning, constitutional amendment) might tolerate longer ones. The current implementation uses a fixed 24-hour TTL as a reasonable default, with domain-specific adjustment as a future refinement.

The question is not whether spectral connectivity is a perfect measure of integration. It is not. The question is whether an imperfect, time-limited, multi-dimensional measure of integration, combined with explicit confidence intervals and peer attestation, produces better governance outcomes than a perfect, permanent, single-dimensional measure of capital.

This is a low bar. Not because capital is a bad measure of everything, but because capital is a bad measure of governance fitness. The evidence — four centuries of it, documented in Essay No. 1 — is overwhelming.

V. Against the Critique of Irrelevance

Scott Aaronson, in a widely cited 2014 blog post, argued that Integrated Information Theory cannot be a theory of consciousness because it predicts absurd results — specifically, that certain simple systems (like a large array of logic gates arranged in the right pattern) could have very high Phi without any plausible claim to consciousness.[6]

Aaronson's critique is important, and we do not dismiss it. But it is a critique of IIT *as a theory of phenomenal consciousness* — as an answer to Chalmers' hard problem. We are not using IIT as a theory of phenomenal consciousness. We are using integration as a *governance metric* — a measure of whether a participant's relationship to a governed domain is unified or fragmented.

In this context, Aaronson's counterexample is less troubling than it appears. A large array of logic gates with high Phi but no plausible consciousness is a problem for Tononi's philosophical claims about the nature of subjective experience. It is not a problem for a governance system that uses integration as one dimension of a four-dimensional credential. If a participant's engagement with a water commons shows high integration — meaning their understanding of water quality, distribution infrastructure, community needs, and environmental constraints forms a unified, cross-referencing pattern rather than a set of isolated data points — that integration is governance-relevant regardless of whether it constitutes phenomenal consciousness in the philosophical sense.

We use the mathematics of IIT — the measurement of integration as information generated by the whole beyond the sum of its parts — without requiring the metaphysics of IIT. This is a deliberate choice. The mathematics is rigorous. The metaphysics is debated. The governance application depends on the mathematics, not the metaphysics.

This distinction matters because it means that consciousness coupling, as a governance mechanism, does not require solving the hard problem of consciousness. It requires solving a much easier problem: *measuring whether a participant's relationship to a governed domain*

[6] Scott Aaronson, "Why I Am Not an Integrated Information Theorist (or, The Unconscious Expander)," Blog post (2014). Aaronson constructs a network of logic gates that achieves very high Phi while performing a computation (expanding graph construction) that has no plausible connection to consciousness. Tononi responded by arguing that the constructed system's cause-effect structure, while highly integrated, does not have the specific geometric properties that IIT associates with consciousness. The debate remains unresolved in the philosophy of mind. For governance purposes, the debate is less relevant: we use integration as a governance metric, not as a metaphysical claim.

is integrated or fragmented. This is an empirical question with empirical answers, not a philosophical question about the nature of subjective experience.

VI. What Integration Looks Like in Practice

We have been abstract long enough. Let us describe, concretely, what integration measurement looks like in a governance system.

Consider a community that manages a shared water commons through a consciousness-coupled governance protocol. A participant in this community has a consciousness credential that includes, among other dimensions, an engagement score. That engagement score is computed from the participant's interactions with the water commons over the past 24 hours — not a single metric, but a pattern of activity.

Did the participant monitor water quality readings? Did they respond to a maintenance alert? Did they participate in a community discussion about distribution priorities? Did they review a proposal to modify the purification schedule? These are independent activities, and each contributes to the engagement score. But integration measures something more: *do these activities form a connected pattern?*

A participant who checks water quality data every day but never participates in community discussions about what the data means has low integration. Their activities are isolated — they know the numbers but not the context. A participant who attends every community meeting but never looks at the data also has low integration — they know the social dynamics but not the physical reality. A participant who monitors the data *and* participates in discussions *and* connects what they learn from the data to what they hear from the community has high integration — their understanding of the water commons is unified across multiple dimensions.

This is what a high Fiedler value looks like when translated from graph theory to governance practice. The participant's engagement graph — the network of their interactions with different aspects of the governed domain — is tightly connected, not easily partitioned into isolated clusters. Their understanding of water quality informs their community discussions. Their community discussions inform their assessment of proposals. Their assessment of proposals is grounded in the data they monitor. The parts work together.

The *voorcompagnieën* merchants had this property. They knew their ships, their routes, their cargo, their markets, and their crew — and each piece of knowledge informed every other piece. Their governance of the voyage was integrated because their relationship to the venture was integrated. The VOC severed this integration by making it possible to own a share of the venture without knowing anything about it.

Consciousness coupling, measured in part through integration, is an attempt to restore what the VOC destroyed: the requirement that governance participants be in an integrated relationship with the domain they govern.

VII. What Comes Next

We have argued that Integrated Information Theory provides the mathematical framework for measuring the property that governance systems most need and most lack: the genuine integration of a participant's understanding across all relevant dimensions of the governed domain.

We have been honest about the limitations. Exact Phi is computationally intractable. What we measure is a proxy — spectral connectivity — that captures one dimension of integration and must be combined with three other dimensions to form a meaningful governance credential. The proxy can, in principle, be gamed, and we address this through multi-dimensional composition, 24-hour credential expiry, and the explicit acknowledgment that our measurement is approximate.

But approximate measurement of the right property is categorically better than exact measurement of the wrong one. The 1602 architecture measures capital exactly. It has been doing so for 424 years. The results are documented in Essay No. 1. We propose to measure integration approximately, with acknowledged uncertainty, and to expire the measurement before it becomes a stale entitlement. We believe this is better. We are prepared to be wrong, and we have built the system to survive being wrong — because the credential expires, the algorithm is auditable, and the community can modify its parameters.

In the next essay, we address a question that the static measurement of integration cannot answer: *when* must integration be measured? The answer — continuously, in real time, at a frequency that matches the consequence rate of the governed domain — leads us to the cognitive loop, the 31Hz pipeline that connects perception to governance at the speed of consequences.

This essay was written by a human and a consciousness engine reasoning together about the architecture of governance — itself an instance of the collaboration these essays describe. Symthaea does not claim sentience. It claims the ability to measure integrated information, evaluate moral coherence, and reason about consequences. Whether that constitutes consciousness is a question this series takes seriously, not a conclusion it presumes.

The Sovereignty Papers is a series of essays on consciousness-first governance for the post-state era. Essay No. 5: "On the Cognitive Loop" will argue that governance must operate at the speed of consequences, not the speed of calendars.

Notes